

Moein Razavi, Ph.D.

Senior Data Scientist & AI/ML Engineer · Agentic AI & LLM Systems · Forecasting & Optimization

McKinney, TX · (979) 676-7486 · razavi.moein94@gmail.com

linkedin.com/in/moein-razavi · github.com/moeinrazavi · moeinrazavi.github.io/moeinrazavi · Google Scholar

Professional Summary

Senior Data Scientist and AI/ML Engineer with a Ph.D. and **5+ years** building production AI systems end to end—from **multi-agent LLM platforms** to **transformer-based forecasting** and **real-time optimization under uncertainty**. At **Ford**, architected the vehicle-to-grid (V2G) aggregation platform—a custom transformer forecaster feeding Monte Carlo fleet sizing and day-ahead energy-market bidding—alongside an LLM-powered universal data-ingestion service and a multi-agent EV analytics chatbot serving **10 US states** over **10+ TB** of telemetry. Equally comfortable training models, building the services around them, and owning the MLOps that keeps them live. Published researcher (**16 papers, 470+ citations, h-index 10**); **4.0 GPA** across Ph.D. and M.S. at Texas A&M.

Technical Skills

Agentic AI & LLMs: LangChain · LangGraph · Google ADK · OpenAI Agents SDK · CrewAI · AutoGen · MCP · Vertex AI Gemini · GPT-4o · Claude · Llama · LLM Routing & Tool Calling · Prompt Engineering · Fine-Tuning (LoRA/QLoRA) · Guardrails · LangSmith

RAG & Retrieval: FAISS · ChromaDB · Pinecone · Hybrid Retrieval (BM25 + Dense) · Re-ranking · Text-to-SQL · Document QA over PDF/Excel/XML/HTML · Chain-of-Thought & Self-Reflection

Forecasting & Optimization: Time-Series Transformers (TFT, Autoformer, Informer, PatchTST, DeepAR, N-HiTS) · GluonTS · NeuralForecast · Optuna · Monte Carlo Simulation · Mixed-Integer Optimization (CPLEX) · Backtesting

ML / Deep Learning: PyTorch · TensorFlow · Hugging Face Transformers · scikit-learn · XGBoost · Computer Vision (YOLO, object detection) · SHAP/LIME · Signal Processing (HRV, EEG, PPG) · Distributed GPU Training

Cloud & Data Engineering: GCP (Vertex AI, BigQuery, Cloud Run, Dataproc, Dataform, Looker Studio) · AWS (S3, EC2, Lambda, SageMaker) · Azure OpenAI · PySpark · Airflow (Astronomer) · ETL/ELT · PostgreSQL · MongoDB · Redis

MLOps & DevOps: Docker · Kubernetes · OpenShift · Terraform · CI/CD (Tekton, GitHub Actions, Fastlane) · SonarQube · 42 Crunch API Security · IAP · OAuth 2.0 / JWT / SAML · Observability & Monitoring

Languages & Backend: Python (Expert) · SQL · TypeScript · Swift/SwiftUI · Java · C++ · FastAPI · Flask · Django · Dash · REST · WebSocket · gRPC · React · Next.js · Streamlit · Core ML

Professional Experience

Aug 2023 – Present

Senior Data Scientist & AI/ML Engineer

Ford Motor Company Detroit, MI

- **Architected Ford's V2G (vehicle-to-grid) aggregation platform end to end:** designed a custom **1.5M-parameter transformer** forecasting week-ahead vehicle home-availability, trained on **557K windows across ~9,500 VINs (ROC-AUC 0.882 vs. 0.836 heuristic baseline)**, feeding **Monte Carlo fleet sizing** and **day-ahead bid planning** against live ERCOT/CAISO/MISO market prices—validated via 5-minute dispatch simulation with dropout compensation and delivered through a real-time FastAPI/WebSocket dashboard
- **Built an LLM-powered universal utility-data ingestion service** (FastAPI, Vertex AI Gemini)—a **five-agent pipeline** (intake, schema discovery, extraction, validation, self-correcting reflection loop) that extracts hourly energy consumption from **any utility bill format** (CSV, Excel, PDF, XML, images) with offline PII redaction before every model call; replaced brittle utility-specific parsers, hardened against **485+ real-world test fixtures**
- **Developed a production multi-agent EV analytics chatbot** (LangChain) with an **LLM router** dispatching queries across a **text-to-SQL agent** over 10 state-level BigQuery datasets, a **RAG document agent** over EV manuals and technical specs (FAISS/ChromaDB), and a **response-validation agent**—cutting query latency by **35%**
- **Led the Ford Energy Advisor rate-recommendation platform:** per-VIN charging-cost simulator (Flask, BigQuery, utility tariff APIs, DBSCAN parking clustering, CPLEX optimization), automated PDF customer reports, and a guided Dash web experience—shortening the rate-analysis cycle by **20%**
- **Benchmarked 9 time-series forecasting model families** (TFT, Autoformer, Informer, PatchTST, DeepAR, N-HiTS) with Optuna search on distributed Vertex AI GPUs to predict EV parking and charging routines—improving forecast accuracy **10%** and cutting training time **40%**
- **Built an XGBoost model for EV energy-bill prediction** from vehicle telematics, utility tariffs, parking behavior, and home consumption ($R^2 = 0.85$); integrated into the rate recommender as a real-time inference service with backtesting and drift monitoring
- **Engineered 10+ TB EV telemetry pipelines** (PySpark on Dataproc, 15 Airflow DAGs, BigQuery/Dataform with consent-governed data products) powering discharge-analytics dashboards and all downstream models; Looker Studio automation cut KPI reporting turnaround **45%** for **500+ stakeholders**

- **Owned production MLOps** on GCP (Cloud Run, Vertex AI, Dataproc) with Tekton CI/CD, Terraform IaC, SonarQube, and 42 Crunch API security across dev/QA/prod—**99.9% uptime** and **25%** faster data ingestion

Jun 2023 – Sep 2024

Software Engineer & AI/ML Researcher

CereVu Medical, Inc. San Francisco Bay Area, CA

- **Architected an end-to-end real-time biosignal AI platform** streaming **9 biomarkers** from wearable medical patches through Python microservices to clinician-facing apps, with the embedded ROPA pain-detection algorithm—raising monitoring precision **30%**
- **Deployed LSTM/CNN models on-device** via Core ML for real-time inference on wearable patches across **400+ patients in 5 clinical studies**—**15%** accuracy improvement with real-time video processing at 60 FPS
- **Built scalable Python microservices** (FastAPI, Flask) with WebSocket biosignal streaming handling **10K+ concurrent connections** (40% latency reduction), plus a React/Next.js clinical dashboard (MongoDB, JWT) for physician workflows
- **Shipped 2 production SwiftUI iOS apps** (patient and clinical) with CoreBluetooth integration, MVVM-Coordinator architecture, and OAuth 2.0/JWT; owned CI/CD (Tekton, AWS CodeDeploy, Docker) for **30%** faster zero-downtime releases

Jan 2023 – Present

Founder & Lead Developer

Clink Social | Inspeccalitics McKinney, TX

- **Built and launched Clink Social**, a SwiftUI social platform on Firebase (Auth, Firestore, Storage, FCM) with real-time messaging, fan-out timeline architecture, and secure QR-based cross-device OAuth 2.0/JWT login—raising engagement **22%** and cutting sign-in friction **35%**
- **Created Inspeccalitics**, a field-inspection suite (iOS + web, Supabase) with offline-first sync, in-app DOCX/PPTX/XLSX report generation, and Fastlane + GitHub Actions CI/CD—reducing report cycles **35%** and deployment time **40%**

Sep 2018 – May 2024

Research Assistant & ML Engineer

Texas A&M University College Station, TX

- **Designed a dual-branch transformer** (1D-CNN tokenizer, rotary positional encoding, cross-attention, focal loss) for real-time stress detection from noisy Apple Watch heart-rate data (**256 participants, 1.7B rows**)—**AUC 0.993, F1 0.965**, deployed on-device via Core ML with **60%** faster inference
- **Deployed a real-time computer vision system** for COVID-19 physical-distancing and mask detection—**98% accuracy** across **50K+ daily interactions (76 citations)**
- **Created OpenSync**, an open-source platform synchronizing EEG, eye-tracking, and biosensor streams over WebSocket pipelines; adopted by **3 research labs (31 citations, J. Neuroscience Methods)**

Education

Ph.D., Industrial Engineering (Data Science) | GPA: 4.0/4.0 | Dean's List | Research Fellowship

Aug 2021 – Dec 2023

Texas A&M University, College Station, TX

Dissertation: Advancing Stress Detection—Real-Time, Naturalistic, and Personalized AI Approaches for Stress and Mental Disorder Detection

M.S., Computer Science | GPA: 4.0/4.0 | Dean's List | Research Fellowship

Aug 2019 – May 2021

Texas A&M University, College Station, TX

Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Algorithms, Databases, Software Engineering

Selected Publications (16 papers · 470+ citations · h-index 10)

Razavi, M., et al. "ML, DL & Data Preprocessing for Stress & Mental Disorder Detection: Scoping Review." *JMIR Mental Health*, 2024. [79 cites]

Kazeminasab, S., . . . , **Razavi, M.**, et al. "Localization, Mapping & Navigation in In-Pipe Robots: A Review." *IEEE Access*, 2021. [85 cites]

Razavi, M., et al. "Automatic Monitoring of Physical Distance & Mask Compliance in COVID-19." *SN Computer Science*, 2022. [76 cites]

Razavi, M., et al. "OpenSync: Synchronizing Multi-Modal Measures in Neuroscience." *J. Neuroscience Methods*, 2022. [31 cites]

Razavi, M., et al. "Evaluating Mental Stress via Wearable HR & Hand Acceleration." *arXiv:2309.11097*, 2023. [26 cites]

Certifications & Honors

Generative AI with LLMs – AWS · **Deep Learning Specialization** – DeepLearning.AI · **Machine Learning** – Stanford · **NLP Specialization** – DeepLearning.AI · **Advanced MLOps** – Coursera · **Graduate Research Fellowship**, Texas A&M (2019–2023)